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ADVANCEMENTS IN CANCER THERAPY THROUGH MIRNA MANIPULATION

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Abstract :

MicroRNAs (miRNAs) are a subclass of short non-coding RNAs used in all cells to control the expression of genes. Typically, only one miRNA is 22 nucleotides long and can control the transcription of many genes. Over the past decade, new opportunities for diagnosis and further therapeutic development have been observed through the study of miRNA biology in cancer. miRNAs are important for phenomena such as genomic instability, transcriptional dysregulation, altered epigenetic regulation, and defects in biosynthetic machinery, and dysregulation of microRNAs is often associated with cancer. Whenever microRNAs change, they can regulate oncogenes and tumor suppressor genes that can influence tumor development. Further studies are essential to confirm the feasibility of such approaches using miRNAs, and their evaluation may open new opportunities to use miRNAs as potential biological cancer markers and therapeutic targets. Our review focuses on how miRNAs control tumor spread and the potential applications of miRNA regulation in cancer therapy. miRNAs can be used as effective therapeutic agents because they tend to target multiple oncogenes and tumor suppressors.

Keywords : Micro RNA Dysregulation, Gene amplification, Cancer Stem Cells

Introduction :

MicroRNAs (miRNAs) are short non-coding RNAs that control the subsequent process of gene declaration. They are typically 20-22 nucleotides long and are fixed in cell 1. Transcriptional activation of oncogenes regulates various biological activities, including cancer. By placing additional regions in the 30 untranslated region (UTR), miRNAs influence a wide range of biological activities through feedback mechanisms such as cell division, proliferation, and death. Because miRNAs are incredibly conserved across all domains of life, it is likely that miRNAs function as broader post-transcriptional regulators [1]. This suggests that they can modulate many genes and signaling pathways, making them attractive targets for therapeutic intervention. Therefore, miRNAs are being investigated as promising tools to treat diseases such as cancer. This study showed that miRNAs play important roles in the regulation of biological processes through precise interactions with proteins, transcription factors, proteins, and RNP complexes. Additionally, mutations in mir-15 and mir-16-1 were found to inhibit cancer growth by inhibiting bcl-2 protein and causing apoptosis. 2 This is important because binding of miRNAs to proteins or transcription factors can change declaration and function.

Additionally, miRNA profiling studies provide new details about the molecular mechanisms of cancer and may serve as a tool for early cancer detection.

miRNAs are being studied for their potential to reduce drug resistance and toxicity and improve the efficiency of existing cancer treatments. Therefore, miRNA-based strategies offer the potential to specifically target specific cancer cells and may be valuable for personalized medicine approaches. Our review focuses on the regulatory mechanisms of miRNAs, biosynthesis, biosynthesis, gene amplification, cancer biomarkers, and cancer stem cells [2].

Materials and methods:

By manipulating the declaration of these proteins, miRNAs can influence the cell cycle, cell survival, and cell differentiation. For example, detected mutations in mir-15 and mir-16-1 inhibit the declaration of the bcl-2 protein, which is known to be involved in regulating apoptosis. Therefore, these mutations may reduce cancer growth by causing apoptosis. The cancer detection, classification, assessment, and predictive capabilities of our miRNAs have been significantly improved through innovations in DNA sequencing and miRNA profiling technologies. A more personalized and precise treatment plan can then be created for each cancer patient. Additionally, miRNA profiling can be used to identify biomarkers for tumor diagnosis and prognosis and to monitor cancer response to treatment. miRNA profiling can also be used to identify potential therapeutic targets and assess the efficacy of potential drugs [3].

miRNA biogenesis, biogenesis and regulatory mechanisms:

miRNAs are developed in a highly consistent manner. miRNAs integrate into the human genome in a variety of ways, including expression from multiple precursors and intergenic transcripts. Each precursor encodes only one strand of premiRNA (later acquiring additional structures such as hairpins).

However, RNA polymerase III also processes some of it. Once the pri-miRNA enters the nucleus, the RNA-binding protein DGCR8 and the type III RNase DROSHA act as endonucleases to cleave the newly synthesized pattern to generate pre-miRNA sequences 80-100 nucleotides long. Masu.

Pre-miRNAs are transported across the plasma membrane to the nucleus by the Ran/GTP/Exportin-5 complex [4]. The pre-miRNA is degraded in the cytoplasm by the cytoplasmic ribonuclease (RNase III) enzyme Dicer into a growing miRNA duplex. This processing technique enhances mature separated-strand miRNAs that bind to Argonaute 2 (AGO2) to form RNA-induced silencing complexes (RISCs).

This complex can essentially bind to the cytosolic mRNA targets of the common 30 UTRs of this complex. Base pair similarity at the 50-terminus of mature miRNAs is the basis for interaction with mRNA. Due to the short length of the interaction site, miRNAs can target a variety of different mRNAs. miRNAs function as receptors in cancer and expand a wide range of communication systems. It has been shown that miRNAs influence Toll-like receptors on natural killer cells by modifying the nuclear factor-kB signaling pathway6-8 [5].

Activity of miRNAs in Cancer:

miRNAs have recently become known as unique cellular elements that behave differently in diseased and healthy cells. The potential of miRNA-based biomarkers for cancer found in various body fluids facilitates diagnosis and monitoring. miRNAs are small non-coding RNAs that can control gene expression and are dysregulated in many disease states, including cancer. Detecting specific miRNAs near a patient's body fluids could potentially point to nearby cancer cells by examining the tumor microenvironment, 9 without the need for biopsies or other invasive procedures. Different types of tumors have distinctive miRNA signatures that help classify 4,444 different types of cancer. To regulate genes, miRNAs mainly play two roles. (1) Homeostatic control of gene regulation [6].

This is highly dependent on cell type. (2) establishment of cell fate and continued maintenance of cell identity via feedback mechanisms; miRNAs act as gene on/off switches and can turn a specific gene on or off by binding to the mRNA encoding that gene in the cell. This allows miRNAs to control the proteins of key molecules involved in the generation and maintenance of the tumor microenvironment, potentially influencing tumor growth and progression. Initiation and inhibition of the AMP-activated protein kinase signaling pathway controls cell survival and Mammalian Target of Rapamycin-Activated Cell Proliferation10. Tumor cells lose the ability to regulate mRNA through microRNAs during mRNA translation due to genetic alterations, single nucleotide polymorphisms, and removal of the 30-UTR. The total amount of mature miRNAs within cells is significantly reduced by polymorphisms that reduce the effectiveness of miRNA interpretation machinery. This can lead to the overexpression of certain mRNAs, which can lead to the growth and spread of cancer cells. miRNA biogenesis and regulation is shown in (figure 1).

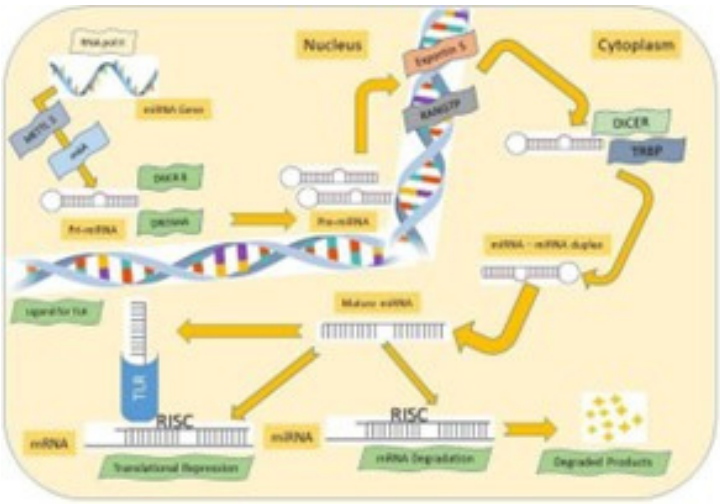

Figure 1: miRNA biosynthesis and regulation.

Gene Amplification/Deletion Affecting miRNAs:

It is believed that miRNA expression levels in cancer cells change, resulting in gene amplification, deletion, or translocation. For example, gene amplification of the miR-17-92 cluster is found in lung cancer and his B-cell lymphoma. Instead, B-cell CLL patients show a decrease in miR-15a/16-1-related genes on chromosome 13q14. The presence of miR-143 and miR-14511-13 in the 5q33 region has been shown in lung cancer patients. Whole-genome sequencing of these samples shows that many miRNA genes are located in tumor-associated genomic regions such as: B. Tumor regulatory genes, oncogenes, or common discontinuous regions. The results show that specific deletions, amplifications, or moved sites within the genome are responsible for changes in miRNA structure [7]. It can disrupt gene regulation and cause cellular transformation and tumor development.

Therefore, understanding the role of miRNAs in this process may lead to the development of more effective treatments for lung cancer. Additionally, understanding how miRNAs are affected by structural changes in the genome may lead to the development of targeted therapies that can restore normal gene regulation and reduce the likelihood of tumor progression. This could lead to more effective treatments with fewer side effects [1].

Incorrect epigenetic modifications:

Cancer cells exhibit atypical epigenetic changes such as: B. Global DNA hypomethylation, variation in histone pattern changes, and hypermethylation of tumor suppressor genes. T24 urinary tract cancer cells exposed to histone acetylation inhibitors and DNA methylation inhibitors showed a >3-fold increase in the expression of 17 miRNAs.

Expression of miR-29 has been shown to be significantly associated with cancer as it inhibits the formation of DNMT3A and DNMT3B, genes, miRNAs, and epigenetic strategies required to control DNA methylation. Therefore, epigenetic strategies such as miRNA expression may be important in the development and progression of urinary cancer [8]. This association between miRNA expression and epigenetic processes is due to the ability of DNMT3A and DNMT3B to control gene methylation, which has been linked to cancer development. The fact that miR-29 expression inhibits the formation of these proteins suggests that miRNAs and epigenetic processes directly influence urinary cancer development and outcome [7].

Discrepancies in miRNA production mechanisms:

Dysfunction of substrates and enzymes in biological pathways such as Dicer, Drosha, DGCR8, and Exportin 5 has a significant impact on overall competent miRNA levels. For example, partial deletion of Dicer1 and Drosha promoted carcinogenesis in various cancer types, as shown in both in vitro and in vivo models. This is because miRNA biogenesis requires multiple enzymatic processes, 5'-phosphate capping, 3'-polyadenylation, and endonucleolytic fragmentation for nuclear export. Dicer and Drosha are responsible for the endonuclease cleavage step. Once dysfunctional, the following steps are disrupted, reducing total levels of competent miRNAs and promoting carcinogenesis. This disruption of the miRNA biogenesis process may lead to increased carcinogenesis rates, so it is important that Dicer and Drosha function properly [9].

Tumors with aberrant miRNA:

Levels The ability to evade growth inhibitors, maintain proliferative signals, increase immortality through recombination, stimulate angiogenesis, and initiate metastatic dissemination are all acquired by tumors. Depending on the gene target, dysregulated miRNAs are thought to act in the form of suppressor genes or oncogenes, affecting one of the traits mentioned above. miRNA profiling of various malignancies shows abnormal levels compared to normal tissues.

Several studies have shown that miRNAs are injected into various cell cycle pathways, thereby sustaining proliferation and avoiding growth slowdown in malignant cells. This allows tumor cells to escape the normal apoptotic pathway and avoid senescence.

Furthermore, tumor cell growth and proliferation is driven by angiogenesis and invasion, which can be regulated by dysregulated miRNAs. Therefore, miRNAs play an important role in cancer development and metastasis, and understanding their behavior is critical to understanding cancer itself [2,8].

Focus on cancer stem cells:

Biological studies have shown that vulnerable sites contain cancer gene-specific miRNAs and are susceptible to miRNA loss or reduction, followed by multiple This leads to the amplification of the cancer gene. These changes influence cancer-related activities such as metastasis, anti-apoptosis, tissue engagement, and therapeutic resilience. A particularly efficient therapy is the cell-based targeted administration of miRNA inhibitors or miRNA mimetics. Therefore, miRNAs serve as physiological indicators of cancer stem cells (CSCs), and further research may reveal the important part of miRNAs in CSCs, especially in diagnosis, prognosis, and treatment. This will improve current cancer treatment protocols and minimize side effects. Additionally, miRNAs can track her CSCs in the body and monitor the effectiveness of cancer treatments. Therefore, understanding the role of miRNAs in cancer can help researchers develop new and more effective treatments. miRNAs offer a unique opportunity to specifically target his CSCs, as they often express miRNAs that are not expressed in normal cells, making them potent adversaries for therapeutic intervention [3]. Targeting these miRNAs could potentially selectively attack and eradicate stem cells without harming normal cells, thereby enabling more effective and safer cancer treatments. As a result, targeting miRNAs may provide a new way to selectively target CSCs and eliminate them without damaging healthy cells, which could lead to more effective and safer cancer treatments [10].

Identification of cancer biomarker miRNA:

microRNA is highly stable in body fluids and is thought to be associated with cancer survivor outcome or treatment response through fluctuations in expression. Circulating miRNAs are released from tumor tissues into the environment and are protected from internal RNase function. This protection allows them to be present in tissues, for example blood samples, and to monitor changes [9]. These expression changes can be used to determine how patients will respond to treatment and predict cancer outcome.

Cancer detection and prognosis are facilitated by the analysis of multiple free-floating miRNAs, which also helps in identifying different tumor groups. Exosomes are enriched with 4,444 microRNAs that confer resistance and are essential for cancer growth and spread21.

By analyzing the different linguistic levels of airborne miRNAs, researchers can determine the type of cancer, its stage, and prognosis. This helps doctors customize treatment plans for each patient and make more informed decisions about treatment. Exosomes also enable a deeper understanding of how cancer spreads, helping to develop new treatments or improve existing ones. Using blood, plasma, and body fluids, many cancer-specific miRNAs encapsulated in exosomes (30-120 nm membrane-derived vesicles) were discovered to quickly and accurately diagnose cancer. A method is provided. For example, has-miR-933 in exosomes produced from gastric fluid is associated with functional dyspepsia22 [11].

miRNA-based medical treatments for cancer:

Tumor suppressor miRNA replication decreases with cancer growth and promotes oncogenic signaling pathways. Therefore, it makes sense to relocate miRNAs that inhibit cancer growth at tumor sites. This can be achieved through the use of miRNA gene therapy, which targets miRNAs against genes involved in cancer proliferation and invasion. miRNA-inspired gene therapy has been shown to effectively suppress tumor growth in animal models. Instead, OncomiR-dependent cancers are controlled using miRNA antagonists (antimiRs) [3]. By targeting stromal cells that support cancer growth, miRNAs also target endothelial cells and fibroblasts, which limit angiogenesis and fibrosis.24 Anti-miRNAs can slow tumor progression by restricting signaling pathways that regulate tumorigenesis, delivering nutrients to tumors, preventing fibrosis, and limiting the growth of new blood vessels that cause further tumor growth can. Therefore, the properties of miRNA guide molecules play an important role in the formation of miRNA antagonists or mimetics. Therefore, targeting miRNA signaling pathways could be an effective and less invasive method to fight cancer [12].

Classification of reasons for intended miRNAs:

Two important factors supporting the application of miRNAs as anticancer agents are that miRNA expression in cancer cells is dysregulated compared to normal tissues, 25 and miRNA Its expression may be targeted to enhance cancer phenotypic changes. miRNAs have the ability to target many genes, usually as part of a network, and can very effectively control a variety of biological functions important for the homeostasis of both healthy and cancer cells.

This makes miRNAs particularly attractive as pharmaceuticals. miR-181 has been shown to be essential for controlling T cell receptor accessibility and signaling strength at the posttranscriptional stage by selectively targeting a panel of phosphatases. Carcinogenesis typically affects the cell cycle, cell adhesion, chromosome stability, and DNA fixation pathways. Because miRNAs control multiple signaling pathways and coordinate their actions in healthy normal cells and tissues, it is logical to hypothesize that they play important functions in regulating malignant networks. miRNAs can be thought of as a "power grid" connecting all these proteins and transcriptional networks. Focusing on specific genes or proteins may not be as effective or comprehensive as developing treatments that alter miRNA production and restore balance [1,7].

miRNA-based therapeutic technologies:

Targeting miRNA activity in cancer involves two major approaches. Direct approaches use oligonucleotide or virus-based structures27 [4]. These techniques can either block miRNA binding or act as miRNA decoys to prevent miRNAs from binding to their targets. Alternatively, indirect techniques utilize antagomirs and miRNA sponges to target miRNA activity28. Indirect approaches modify miRNA expression using drugs that focus on miRNA synthesis and processing.

Barriers to miRNA-based interventions:

The need for tissue-specific distribution and cellular uptake of synthetic oligonucleotides sufficient to enable sustained target blockade poses challenges when applying miRNA therapy in vivo. One of the biggest challenges. Due to lack of safety and efficiency, it is not suitable for in vivo applications. Therefore, further development of new delivery systems is required to successfully apply miRNA therapy in vivo. The biological instability of these materials in human body fluids or tissues is the first hurdle to overcome, as unchanged "naked" oligonucleotides are rapidly degraded by cellular and serum nucleases. Researchers have investigated delivery systems such as cationic liposomes, micelles, polymers, and peptides to address this issue. These protect and stabilize the oligonucleotide while allowing it to penetrate into target cells. Moreover, these delivery systems also have the potential to specifically target certain cell types, thereby increasing the efficiency of therapy [8]. However, oligonucleotides have opposite charges and are larger. The second barrier, low absorption into cells, can make it difficult to penetrate cells [7,9]. To counter this, oligonucleotide modifications are used. miRNA sponges are transcripts generated from mammalian expression vectors with multiple critical tandem binding sites for specific miRNAs.

The miRNA sponge moves to the miRNA of interest and prevents it from associating with its target mRNA, thereby reducing the expression of the intercepted gene29. This approach allows for more targeted modulation of miRNA frequencies, a more effective strategy than direct miRNA inhibition. miRNA sponges can be designed to target specific miRNAs with high affinity, prevent their binding to intercepting mRNAs, and reduce the amount of intercepted genes. This method has advantages over direct miRNA inhibition as it is more precise and can be used to control specific gene expression pathways. Additionally, miRNA sponges can be engineered to achieve higher levels of miRNA knockdown and further improve the efficacy of miRNA regulation. Contains cationic lipids and polyethylene glycol and was developed to enhance delivery to cells [2,6].

Conclusion:

Subsequently, miRNA deletion was found to be associated with his CLL. Researchers around the world have studied the function of miRNAs in different types of cancer and why miRNA expression becomes dysregulated. This has revealed the function of many processes, including changes in miRNA synthesis, epigenetic factors, aberrant regulation of gene transcription, and deletion and amplification of miRNA genes. Although miRNAs have a wide range of potential targets, they are thought to cause tumors by altering specific targets and acting as oncogenes or tumor suppressors. Several miRNA inhibitors and miRNA mimetic are being studied in the clinic and could potentially be used therapeutically. Additionally, the functions of non-coding RNAs, such as circular non-coding RNAs and long non-coding RNAs, have also been studied.

Conflict of interest:

Author declares no conflict of interest.

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